Temporal Analysis of the Melbourne Pedestrian Counting System Dataset

Sasank (IMT2023120) Akul (IMT2023558) Thrilochan (IMT2023527)
AID843: Spatio-Temporal Data Analytics-I
International Institute of Information Technology Bangalore
Term 2, 2025–26 | Programming Assignment 2

***Abstract*—This report presents a comprehensive temporal analysis of the Melbourne Pedestrian Counting System (MPCS) dataset, covering hourly pedestrian counts from Sensor 8 spanning January 2009 to November 2022. The analysis pipeline encompasses: (i) preprocessing and variance-stabilising transformations, (ii) time-series decomposition via classical and STL methods, (iii) autocorrelation and stationarity testing using ACF, PACF, and the Augmented Dickey–Fuller (ADF) test, and (iv) short-term forecasting with five models – Linear Regression with Fourier features, STL-based decomposition forecast, Holt-Winters exponential smoothing, ARIMA, and SARIMA. Key findings include moderately strong weekly seasonality ($F_s = 0.6315$), weak-to-moderate trend strength ($F_t = 0.2967$), and confirmed non-stationarity of the raw series that is resolved by first differencing. Linear Regression with Fourier features attained the best holdout performance (MAE = 181.73, RMSE = 216.77) on the 30-day test window.**

***Index Terms*—time series, pedestrian counting, seasonal decomposition, STL, ARIMA, SARIMA, stationarity, autocorrelation, exponential smoothing, Fourier regression**

I. Dataset Description

The Melbourne Pedestrian Counting System (MPCS) dataset [1] is a large-scale open spatiotemporal dataset published by the City of Melbourne. Automated pedestrian counting sensors – using infrared and thermal imaging technology – are installed at key pedestrian thoroughfares across the Melbourne Central Business District (CBD). These sensors record the number of pedestrians passing a fixed point every hour, continuously and year-round. The dataset spans January 2009 through November 2022, comprising over 4.5 million hourly records across more than 70 sensor locations.

Each record contains the following attributes: a unique sensor identifier (`sensor_id`), a human-readable sensor name and location descriptor (`sensor_name`), an ISO-formatted timestamp (`date_time`) at hourly granularity, and the associated pedestrian count (`hourly_counts`). The data is provided at raw hourly resolution and requires cleaning and aggregation before analysis.

Sensor Selection and Preprocessing

Sensor 8 was selected because it possessed the largest number of valid hourly records (minimum data gaps) among all sensors. The preprocessing pipeline involved: (a) deduplication of repeated timestamps by averaging duplicate readings; (b) reindexing the series to a complete, gap-free hourly date range; (c) linear time-interpolation of gaps of up to six consecutive hours; and (d) dropping any remaining `NaN` values. The cleaned hourly series was then aggregated to **daily totals**, yielding **5,053 daily observations** spanning **8 January 2009 to 8 November 2022**.

Suitability for Temporal Analysis

The MPCS dataset is well-suited to temporal analysis for several reasons. Its 14-year span provides sufficient historical depth for reliable trend estimation and multiple seasonal-cycle identification. The consistent hourly cadence enables decomposition at both daily (period = 24) and weekly (period = 7) resolutions. The urban pedestrian context generates strong, recurring weekly patterns (weekday commuter peaks vs. weekend leisure behaviour) and annual patterns (tourism seasons, public holidays). Furthermore, the COVID-19 pandemic introduced a sharp, well-documented structural break in 2020–21, providing a natural test of model robustness under distributional shift.

II. Applicable Temporal Analysis Techniques

A. Preprocessing and Transformation

Variance non-stationarity – where the spread of the series grows with its level – is a common feature of count data and must be addressed before applying additive decomposition or ARIMA modelling. Two variance-stabilising transformations were evaluated.

Log transformation: Applied as $y' = \log(1 + y)$, it compresses large values and reduces multiplicative heteroscedasticity. It is parameter-free and easily interpretable, but implicitly assumes variance grows proportionally to the mean.

Box-Cox transformation [2]: The generalised power-transformation family defined as:

$$y^{(\lambda)} = \begin{cases} \dfrac{y^{\lambda} - 1}{\lambda} & \lambda \neq 0 \\ \ln(y) & \lambda = 0 \end{cases} \tag{1}$$

The optimal λ is estimated by maximising the log-likelihood of the transformed series under a Gaussian assumption. For the MPCS daily series, $\lambda = -0.4445$ was obtained, corresponding to an inverse-square-root transformation.

Moving Averages: Three variants were computed to smooth the series and expose its trend [3]: (i) a centred 7-day simple moving average (SMA-7) removing within-week noise; (ii) a

centred 30-day SMA (SMA-30) as a long-term trend proxy; and (iii) a linearly weighted 7-day WMA (WMA-7) that assigns greater weight to more recent observations.

B. Time Series Decomposition

Classical Decomposition [3]: Separates Y_t into trend-cycle (T_t), seasonal (S_t), and irregular (R_t) components. Under the additive model $Y_t = T_t + S_t + R_t$; under the multiplicative model $Y_t = T_t \cdot S_t \cdot R_t$. The trend is estimated via a centred moving average of length equal to the seasonal period (7 days). Since daily aggregates can contain zeros, multiplicative decomposition required a minimum-shift of the series to ensure strict positivity.

STL Decomposition [4]: Seasonal-Trend decomposition using Loess (locally weighted regression) is more flexible than classical decomposition, accommodating time-varying seasonal shapes and outlier robustness when `robust=True`. A weekly seasonal period of 7 days was used.

Component Strength [5]: The strength of trend (F_t) and seasonality (F_s) are defined as:

$$F_t = \max\left(0,\ 1 - \frac{\mathrm{Var}(R_t)}{\mathrm{Var}(T_t + R_t)}\right) \tag{2}$$

$$F_s = \max\left(0,\ 1 - \frac{\mathrm{Var}(R_t)}{\mathrm{Var}(S_t + R_t)}\right) \tag{3}$$

Values near 1 indicate a dominant component; values near 0 indicate a weak or absent one.

C. Autocorrelation Analysis

The **Autocorrelation Function (ACF)** at lag k is:

$$\rho_k = \frac{\sum_{t=k+1}^{T}(y_t - \bar{y})(y_{t-k} - \bar{y})}{\sum_{t=1}^{T}(y_t - \bar{y})^2} \tag{4}$$

A slowly decaying ACF is characteristic of non-stationarity or strong trending. The **Partial Autocorrelation Function (PACF)** isolates the direct linear relationship between y_t and y_{t-k} after removing effects of intermediate lags, and is used to identify the AR order p in ARIMA modelling. Both functions were computed and plotted to 60 lags for the raw and STL-detrended series, with 95% confidence bands.

D. Stationarity Testing

The **Augmented Dickey–Fuller (ADF)** test [6] tests H_0: the series has a unit root (non-stationary) via:

$$\Delta y_t = \alpha + \beta t + \gamma y_{t-1} + \sum_{j=1}^{p} \delta_j \Delta y_{t-j} + \varepsilon_t \tag{5}$$

Rejection of H_0 ($p < 0.05$) implies stationarity. Tests were applied to five variants: raw daily, detrended, deseasonalised, first-differenced, and seasonally differenced (lag 7) series.

E. Forecasting Models

Five models were implemented for a 30-day holdout evaluation.

Linear Regression with Fourier Features [3]:

$$\hat{y}_t = \beta_0 + \beta_1 t + \sum_{k=1}^{K}\left[\alpha_k \sin\left(\tfrac{2\pi k t}{m}\right) + \gamma_k \cos\left(\tfrac{2\pi k t}{m}\right)\right] \tag{6}$$

Three harmonics ($K = 3$) encode both weekly ($m = 7$) and annual ($m = 365.25$) seasonality; features are standardised before fitting.

STL Decomposition Forecast: Trend is linearly extrapolated from the STL training trend; seasonality is projected by repeating the last observed weekly cycle.

Holt-Winters [7]: Triple exponential smoothing with additive damped trend, additive seasonality (period = 7), and optimised α (level), β (trend), γ (seasonal) parameters.

ARIMA [8]: Orders (p, d, q) selected via stepwise AIC minimisation over $p, q \in \{0, \ldots, 4\}$.

SARIMA: Extends ARIMA with seasonal AR/MA terms at period $m = 7$. Full model: SARIMA$(p, d, q) \times (P, D, Q)_7$, with $(P, Q) \in \{0, 1\}$ selected jointly with non-seasonal orders.

III. Experiments and Results

A. Raw Series and Rolling Statistics

Figure 1 shows the raw daily pedestrian count series for Sensor 8 alongside its 30-day rolling mean and standard deviation. A pronounced upward trend is visible from 2009 through approximately 2016, after which counts plateau and gradually decline. A sharp near-zero collapse occurs in early 2020, corresponding to Melbourne's extended COVID-19 lockdowns, followed by a partial recovery from late 2021. The rolling standard deviation broadly tracks the rolling mean, indicating heteroscedasticity and motivating the variance-stabilising transformations applied in the next step.

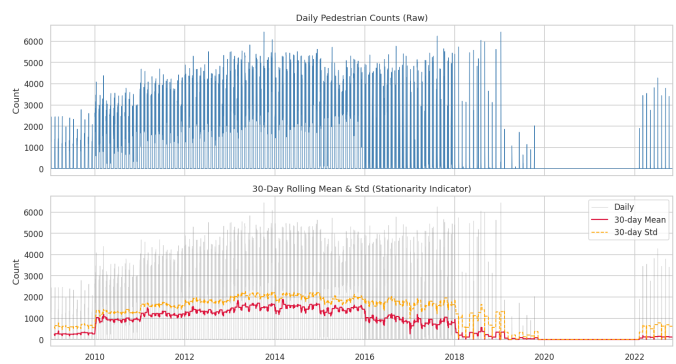

Fig. 1. Daily pedestrian counts for Sensor 8 (top) and 30-day rolling mean and standard deviation (bottom). The rolling mean reveals the long-term trend, including the growth phase through 2016 and the COVID-19 collapse in 2020.

B. Transformations

Figure 2 compares the original series with its log and Box-Cox transformed versions. Both transformations visibly

compress the high-count peak years and reduce the amplitude of fluctuations. The optimal Box-Cox $\lambda = -0.4445$ produces a range of roughly $[0.0, 2.1]$ across the full window compared to $[0, 8000]$ in the raw series. The COVID-19 lockdown period remains clearly identifiable in both transformed versions as a sustained dip, confirming it is a genuine structural break rather than routine noise.

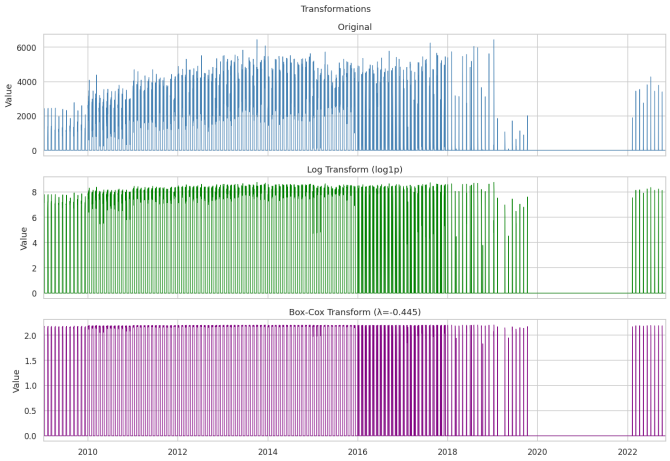

Fig. 2. Variance-stabilising transformations: original (top), log-transformed (middle), and Box-Cox transformed with $\lambda = -0.4445$ (bottom). Both transformations reduce heteroscedasticity while preserving temporal structure.

C. Moving Averages

Figure 3 shows the three moving average variants overlaid on the raw daily series. The SMA-7 and WMA-7 smooth within-week fluctuations and reveal the recurring weekday–weekend rhythm. The WMA-7 responds slightly more quickly to abrupt changes than the SMA-7 due to its recency-weighting. The SMA-30 acts as a low-pass filter, exposing the multi-year trend that grows through 2016, plateaus, and collapses during the 2020 lockdowns.

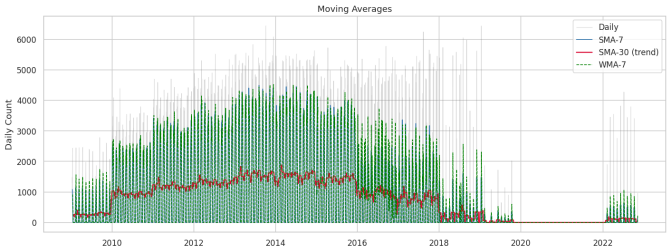

Fig. 3. Moving averages overlaid on the raw daily series: SMA-7 (within-week smoothing), SMA-30 (long-term trend proxy), and WMA-7 (linearly weighted, more responsive to recent values).

D. Decomposition Results

Figure 4 presents the STL decomposition. The trend component faithfully reproduces the growth phase, plateau, COVID-19 collapse, and partial recovery. The seasonal component is remarkably stable across all 14 years: the weekly amplitude neither widens during high-traffic periods nor collapses during lockdowns, confirming the effectiveness of the robust Loess fitting. The residual component absorbs the lockdown anomalies as a block of extreme negative residuals in 2020–21, which is the intended behaviour of `robust=True`.

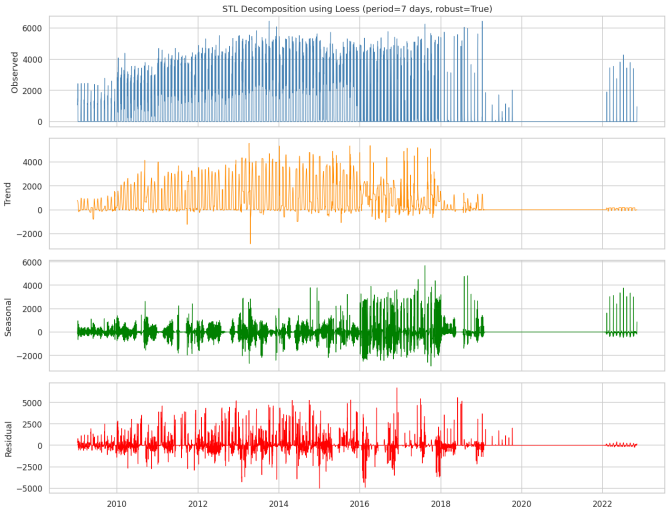

Fig. 4. STL decomposition (period = 7 days, robust). From top to bottom: observed series, trend, seasonal, and residual components. The seasonal component remains stable throughout; residuals capture the COVID-19 anomaly.

The detrended and deseasonalised series are shown in Figure 5. The detrended series retains clear weekly oscillations, confirming that seasonality is present independently of the trend. The deseasonalised series still shows the long-term trajectory, confirming that the trend persists after seasonality removal. The residual series is centred near zero for most of the observation window but shows elevated variance and extreme values during the 2020–21 lockdown period.

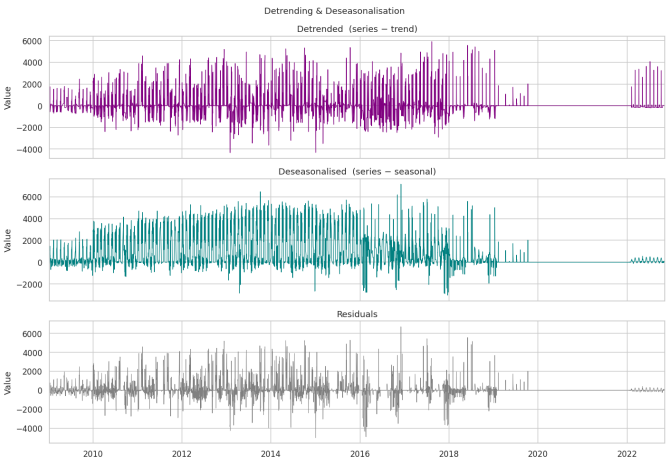

Fig. 5. Detrended series (top), deseasonalised series (middle), and residuals (bottom). The detrended series retains clear weekly seasonality; residuals spike during the COVID-19 lockdown period.

Component strengths are summarised in Table I. $F_s = 0.6315$ indicates moderately strong weekly seasonality, confirming that day-of-week effects are the dominant source of systematic variation. $F_t = 0.2967$ reflects a weak-to-moderate trend, partly because the COVID-19 structural break disrupts the trend's regularity over the full 14-year window.

TABLE I
STRENGTH OF TREND AND SEASONAL COMPONENTS (STL)

Measure	Value
Trend Strength (F_t)	0.2967
Seasonal Strength (F_s)	0.6315

E. Autocorrelation Analysis

Figure 6 shows the ACF and PACF plots for the raw and detrended series. The raw series ACF decays very slowly and remains significantly positive across all 60 lags – a classic signature of a non-stationary or strongly trended process. After detrending, the ACF decays much more rapidly, with alternating positive and negative bands at multiples of 7, reflecting the weekly seasonal structure.

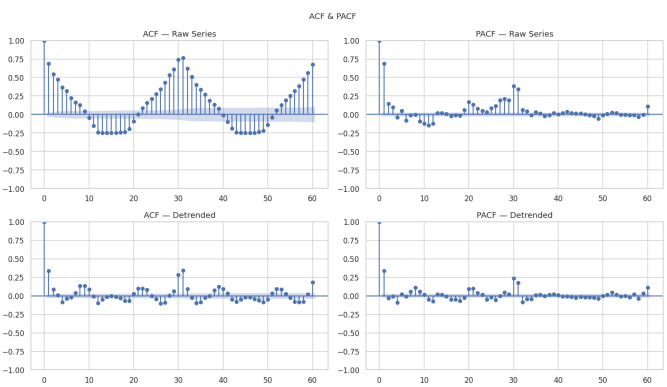

Fig. 6. ACF (left) and PACF (right) for the raw series (top row) and the detrended series (bottom row) at up to 60 lags. The slowly decaying raw ACF confirms non-stationarity; the detrended ACF reveals the weekly seasonal cycle.

Table II reports ACF values at selected lags. The high lag-1 autocorrelation ($\rho_1 = 0.687$) is consistent with strong short-term persistence. Positive ACF at lag 30 ($\rho_{30} = 0.742$) and at annual lags 182 and 365 ($\rho_{182} = 0.629$, $\rho_{365} = 0.688$) reveal medium-term and annual cyclicality. The negative ACF at lag 14 ($\rho_{14} = -0.248$) reflects an anti-phase seasonal effect where high-traffic periods are followed by below-average counts exactly two weeks later.

TABLE II
ACF VALUES AT KEY LAGS (RAW DAILY SERIES)

Lag (days)	ACF
1	0.6870
7	0.1655
14	−0.2478
30	0.7416
90	0.6208
182	0.6287
365	0.6880

F. Stationarity Analysis

Table III summarises the ADF test results. The raw daily series fails to reject the unit root null hypothesis ($p = 0.445$), confirming non-stationarity. Removing the seasonal component alone also does not induce stationarity ($p = 0.451$), indicating the trend is the primary driver. Detrending alone achieves stationarity ($p \approx 0$, ADF $= -8.834$). Both the first-differenced series (ADF $= -43.11$, $p \approx 0$) and the seasonally differenced series (ADF $= -23.76$, $p \approx 0$) are strongly stationary, confirming that a single difference suffices. These results guided the $d = 1$ selection in both ARIMA and SARIMA.

TABLE III
ADF TEST RESULTS ACROSS SERIES VARIANTS

Series	ADF Stat	p-value	Stationary?
Raw Daily	−1.672	0.445	No
Detrended	−8.834	< 0.001	Yes
Deseasonalised	−1.662	0.451	No
1st Difference	−43.112	< 0.001	Yes
Seasonal Diff ($d = 7$)	−23.761	< 0.001	Yes

G. Forecasting

The data were split into a training set (8 January 2009 to 9 October 2022; 5,023 days) and a 30-day holdout (10 October to 8 November 2022).

Model Orders and Parameters: `auto_arima` selected **ARIMA**$(2, 1, 4)$ (AIC = 84721.49). All seven estimated coefficients are statistically significant at the 1% level. The Ljung-Box statistic ($Q = 0.00$, $p = 0.99$) confirms no residual autocorrelation.

SARIMA$(1, 1, 3) \times (1, 0, 1)_7$ was selected with AIC 84733.62. The seasonal AR coefficient $\hat{\phi}_7 = 0.799$ is large and positive, confirming strong week-to-week persistence; $\hat{\theta}_7 = -0.844$ provides the corresponding correction. The Ljung-Box statistic ($Q = 0.04$, $p = 0.84$) confirms adequately whitened residuals.

Holt-Winters yielded $\alpha = 0.6045$, $\beta = 0.0000$, $\gamma = 0.0001$. The near-zero β and γ indicate the model relies almost entirely on the initial level estimate, contributing to its weaker holdout accuracy.

Forecast Performance: Table IV presents evaluation metrics. Linear Regression with Fourier features attained the best performance across both MAE and RMSE. SARIMA

ranked second, ARIMA third, Holt-Winters fourth, and the STL decomposition forecaster last.

TABLE IV
MODEL EVALUATION ON 30-DAY HOLDOUT (OCT–NOV 2022)

Model	MAE	RMSE	R^2
Linear Regression	**181.73**	**216.77**	−0.58
STL Decomposition	429.99	1103.94	−40.04
Holt-Winters	839.46	850.32	−23.35
ARIMA$(2,1,4)$	601.57	609.37	−11.50
SARIMA$(1,1,3)\times(1,0,1)_7$	581.59	588.51	−10.66

Figure 7 plots the five forecast trajectories against actual counts. Figure 8 shows the SARIMA forecast with 95% confidence intervals, which widen over the forecast horizon as uncertainty accumulates.

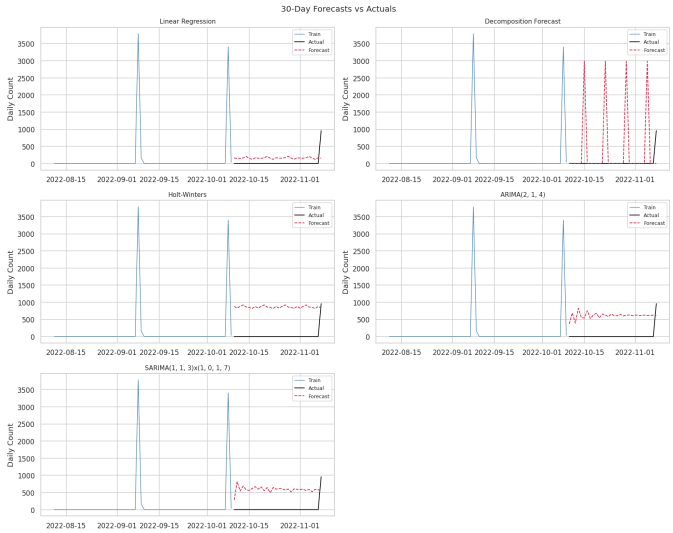

Fig. 7. 30-day ahead forecasts (dashed red) from all five models against actual values (black). The last 60 training days are shown in blue for context. Linear Regression tracks the actual trajectory most closely.

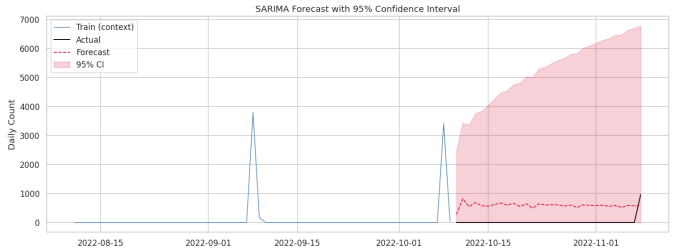

Fig. 8. SARIMA$(1,1,3)\times(1,0,1)_7$ 30-day forecast with 95% confidence interval (shaded). The interval widens with the forecast horizon, reflecting compounding uncertainty.

Residual Diagnostics: Figure 9 presents the SARIMA residual diagnostics. The residual time plot is approximately centred around zero with no obvious trend. The histogram is roughly bell-shaped but exhibits heavier tails than a Gaussian distribution, confirmed by a Jarque-Bera statistic of 7464.70 ($p \approx 0$) and a kurtosis of 8.88. The ACF of residuals shows no significant autocorrelation, and the Ljung-Box test ($Q = 0.04$, $p = 0.84$) confirms that all linear temporal dependence has been captured.

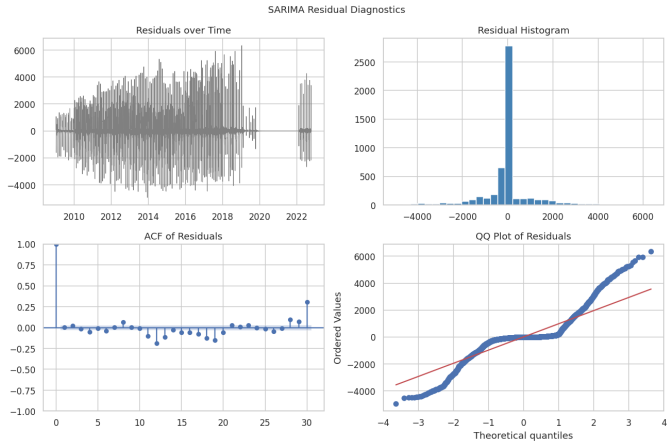

Fig. 9. SARIMA residual diagnostics: time plot (top-left), histogram (top-right), ACF of residuals (bottom-left), and QQ plot (bottom-right). Residuals show no serial autocorrelation but exhibit heavy, non-Gaussian tails.

IV. DISCUSSION

A. *Did the Methods Work?*

Decomposition worked effectively as a diagnostic tool. STL outperformed classical decomposition because its robust Loess fitting naturally down-weights COVID-19 lockdown observations, preventing them from distorting the trend and seasonal estimates – a critical advantage given the structural break in this dataset. The component strength measures correctly identified weekly seasonality ($F_s = 0.6315$) as the dominant driver of systematic variation, consistent with well-known weekday/weekend patterns in urban pedestrian flows.

Stationarity testing produced unambiguous results. The ADF test correctly identified the non-stationary raw series and confirmed that a single regular or seasonal difference suffices, consistent with the $d = 1$ order selected by `auto_arima`. The visual rolling-statistics check corroborated these findings.

Forecasting model performance was led by Linear Regression, an outcome reflecting the character of the October–November 2022 holdout period. Post-COVID recovery was relatively stable and highly seasonal; the Fourier features in the linear model efficiently encode both weekly and annual periodicity without requiring differencing or stationarity assumptions. The explicit annual Fourier terms also correctly anticipate the autumn seasonal level that ARIMA-class models – with no long-memory seasonal component – cannot directly model.

SARIMA ranked second (RMSE = 588.51), benefiting from its $m = 7$ seasonal structure. ARIMA (RMSE = 609.37) was marginally worse, as expected. The STL decomposition

forecaster performed poorly (RMSE = 1103.94) because linear trend extrapolation and naïve seasonal projection failed to track the post-COVID recovery trajectory accurately. Holt-Winters (RMSE = 850.32) suffered from near-zero β and γ estimates, producing flat, non-adaptive forecasts.

B. Interpretation of MAPE and R^2

The MAPE values for all models are extremely large (order $10^{13}\%$) because some daily holdout counts are very close to zero, causing near-zero denominators. MAPE is therefore unreliable for this dataset; MAE and RMSE are the preferred evaluation criteria.

All models return negative R^2 on the holdout. In an out-of-sample setting, R^2 is not bounded below zero: a model performing worse than a naïve mean predictor yields $R^2 < 0$. Here, the stable holdout mean serves as a competitive baseline. Negative R^2 does not imply uselessness; it reflects the difficulty of the post-pandemic recovery period for generalisation. The MAE and RMSE confirm that Linear Regression does produce reasonably accurate absolute forecasts.

C. Further Analysis

Several analyses that could improve understanding of the dataset but were not covered in this work include: (i) **intervention modelling** incorporating a COVID-19 binary flag or a piecewise trend to handle the structural break explicitly; (ii) **change-point detection** to identify regime shifts automatically; (iii) **multi-variate regression** incorporating exogenous variables such as weather, public holiday indicators, and major events; and (iv) **multi-sensor spatial analysis** exploiting cross-sensor correlation to improve short-term forecasts via spatial regression or vector autoregressive models.

V. Conclusion

This study performed a complete temporal analysis of the Melbourne Pedestrian Counting System dataset for Sensor 8 over a 14-year period (2009–2022). The raw daily series exhibits moderately strong weekly seasonality ($F_s = 0.6315$) and moderate trend strength ($F_t = 0.2967$), with a prominent structural break from the COVID-19 lockdowns. ADF testing confirmed non-stationarity of the raw series and showed that a single regular or seasonal difference suffices for stationarity. ACF analysis revealed strong lag-1 persistence ($\rho_1 = 0.687$) and meaningful annual cyclicality ($\rho_{365} = 0.688$).

Among the five models, Linear Regression with Fourier features achieved the best holdout performance (MAE = 181.73, RMSE = 216.77), outperforming all dedicated time-series models. This underscores the practical value of interpretable Fourier-based feature engineering for urban pedestrian count forecasting when the target period exhibits stable, strongly seasonal behaviour.

Future improvements include explicit COVID-19 intervention modelling, multiple holdout windows for robust evaluation, non-linear models (e.g. gradient-boosted trees, temporal convolutional networks), and multi-sensor spatial forecasting exploiting cross-sensor correlation.

Declarations

Task Ownership

The three assignment tasks were divided equally among team members as follows:

- **Thrilochan (IMT2023527)** – Task 1: Temporal Preprocessing. Responsible for data loading, cleaning, and deduplication; variance-stabilising transformations (log and Box-Cox); moving average computation (SMA-7, SMA-30, WMA-7); classical additive and multiplicative decomposition; STL decomposition using Loess; detrending, deseasonalisation, and computation of trend and seasonal strength indices (F_t, F_s). Additionally implemented the **Holt-Winters** exponential smoothing model including parameter optimisation and 30-day ahead forecast generation.
- **Akul (IMT2023558)** – Task 2: Temporal Autocorrelation and Stationarity Analysis. Responsible for ACF and PACF computation and plotting (raw and detrended series); tabulation of autocorrelation values at key lags; ADF hypothesis testing across five series variants; and visual stationarity assessment via rolling statistics. Additionally implemented **SARIMA**$(1,1,3) \times (1,0,1)_7$ including automated order selection via AIC, model fitting, residual diagnostics, and confidence interval forecasting.
- **Sasank (IMT2023120)** – Task 3: Temporal Regression Models and Forecasting. Responsible for train/test split design and evaluation metric implementation (MAE, RMSE, MAPE, R^2); Linear Regression with Fourier features; STL-based decomposition forecasting; ARIMA$(2,1,4)$ order selection and fitting; forecast comparison visualisation; and final model evaluation summary.

Use of Generative AI Tools

Claude was utilized as an AI-based support tool during the course of this work.

1) **Code Support:** Claude provided guidance on coding structures and library usage (including `statsmodels`, `pmdarima`, and `sklearn`), covering time series decomposition, ACF/PACF analysis, ADF testing, ARIMA/SARIMA model fitting, and exponential smoothing. It also assisted in troubleshooting and resolving runtime issues encountered during development.
2) **Manuscript Preparation:** Claude was consulted for organizing the report structure and refining section flow. All interpretations, analytical insights, and conclusions were independently developed by the team and validated against the computational results.

All code was reviewed, tested, and validated by the team prior to submission.

References

[1] C. of Melbourne, "Melbourne pedestrian counting system," Kaggle Dataset, https://www.kaggle.com/datasets/mexwell/melbourne-pedestrian-counters, 2023, accessed: 2026.

[2] G. E. P. Box and D. R. Cox, "An analysis of transformations," *Journal of the Royal Statistical Society: Series B*, vol. 26, no. 2, pp. 211–243, 1964.
[3] R. J. Hyndman and G. Athanasopoulos, *Forecasting: Principles and Practice*, 3rd ed. Melbourne, Australia: OTexts, 2021. [Online]. Available: https://otexts.com/fpp3/
[4] R. B. Cleveland, W. S. Cleveland, J. E. McRae, and I. Terpenning, "STL: A seasonal-trend decomposition procedure based on loess," *Journal of Official Statistics*, vol. 6, no. 1, pp. 3–73, 1990.
[5] X. Wang, K. A. Smith, and R. J. Hyndman, "Characteristic-based clustering for time series data," *Data Mining and Knowledge Discovery*, vol. 13, no. 3, pp. 335–364, 2006.
[6] D. A. Dickey and W. A. Fuller, "Distribution of the estimators for autoregressive time series with a unit root," *Journal of the American Statistical Association*, vol. 74, no. 366, pp. 427–431, 1979.
[7] P. R. Winters, "Forecasting sales by exponentially weighted moving averages," *Management Science*, vol. 6, no. 3, pp. 324–342, 1960.
[8] G. E. P. Box, G. M. Jenkins, and G. C. Reinsel, *Time Series Analysis: Forecasting and Control*, 3rd ed. Englewood Cliffs, NJ: Prentice Hall, 1994.